

Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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ABSTRACT

We introduce Stablecoin StressBench, a reproducible, execution-aware benchmark for stablecoin dislocation models. The benchmark provides: an 18-event stress taxonomy across seven mechanism classes, per-minute execution-aware labels that separate quoted price dislocations from net-profitable arbitrage after order-book depth, VWAP slippage, and trading costs, a hindsight oracle defining the performance ceiling, and a model ladder for future work. The central finding is a 12× price-to-execution gap in the March 2023 USDC/SVB stress window: 34.3% of 1-minute windows exceed a 10bps max cross-quote basis on price alone, while only 2.88% remain net-profitable at \$10K notional after costs. A hindsight oracle earns +162bps per trade; on executable arbitrage tasks every active model loses money, and the oracle gap exceeds 260bps. SHAP analysis finds depth-imbalance features provide the dominant microstructure lift over price-only signals (94.8% price vs. 3.4% book/frag). A GRU trained on 30-minute order-book windows partially closes the oracle gap (+36.7bps vs. oracle’s +162.2bps). This reframes stablecoin arbitrage prediction as an execution-barrier problem and provides infrastructure for future work on execution policy design and optimal execution research.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 INTRODUCTION

Stablecoins promise \$1 parity but stress events repeatedly fragment prices across venues, creating apparent arbitrage. A quoted price gap is not a trade: a trader must clear order-book depth, absorb VWAP slippage, pay fees, and bear settlement risk. These frictions are small individually but large jointly, making many stablecoin dislocations optical rather than executable.

This paper introduces Stablecoin StressBench, which separates three layers: optical dislocations, executable arbitrage, and ex-ante predictable executable opportunities. During the March 2023 USDC/SVB event, 34.3% of minutes show a > 10bps basis on price alone; only 2.88% remain net-profitable after VWAP execution at \$10K notional — a 12×

gap. Every active model loses money on executable arbitrage tasks. The oracle gap exceeds 260bps.

Contributions.

- (1) Execution-aware benchmark: per-minute labels with documented order-book depth provenance, VWAP book-walking, fees, notional scaling, and a hindsight oracle ceiling.
- (2) 18-event stress taxonomy: seven mechanism classes (2020–2023) with data-tier governance of permitted claims. Terra/LUNA cross-mechanism check gives 5.9× vs. the Tier-A 12× ratio.
- (3) Oracle gap evidence: +162bps oracle vs. +17.2bps best model (basis task), > 260bps gap on executable arbitrage — structural, not label-mismatch.
- (4) Model ladder, diagnostics, and a partial positive result: SHAP shows depth-imbalance dominates microstructure lift; a GRU on 30-min order-book windows achieves +36.7bps — partially closing the 145bps oracle gap. An oracle-gap decomposition identifies timing as the dominant challenge for future models.

Optimal execution interpretation. The oracle gap has a natural decision-theoretic framing: the state at each minute is the current order book and cross-quote basis, the action is whether to execute the arbitrage route, and the reward is the net bps captured. The oracle corresponds to the optimal policy under perfect foresight. StressBench therefore provides a reproducible environment for learned execution research: the state space is the four-level microstructure feature hierarchy, the reward function is the documented net-profit label, and the performance ceiling is the oracle gap.

2 RELATED WORK

Stablecoin stress. The USDC/SVB episode produced well-documented price fragmentation; we address ex-ante predictability. Uhlig [1] analyses Terra/LUNA mechanics (our validation split); Griffin and Shams [2] study Tether manipulation (our focus is execution microstructure).

Limits to arbitrage and crypto fragmentation. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish theory. Makarov and Schoar [8] show persistent cross-venue crypto price gaps that are not executable profits. Hautsch et al. [3] show settlement latency is a first-order arbitrage cost; Cont and Kukanov [6] show VWAP price—not quoted price—is the relevant object.

Financial ML and benchmarks. López de Prado [10] introduces meta-labeling; Adams and MacKay [17] introduce BOCPD. Cintra and Holloway [11] detect the Terra depeg 12 hours early via Curve AMM pool reserves. We find BOCPD achieves only AUROC 0.23 on CEX cross-quote data — because AMM reserves accumulate gradually while

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy [†]	real L2 (futures)
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell BTC-USDT	real L2 (futures)	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

[†]BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits. The 2.88% positive rate implies severe class imbalance; AUROC is reported but economic net bps is the primary criterion.

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

CEX basis is mean-reverting, making changepoint detection unreliable on quoted prices. To our knowledge, no prior benchmark provides execution-aware stablecoin labels with route-depth provenance and systematic price-to-execution gap analysis.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The gold-layer dataset contains 56,134 1-minute bars across six windows (Table 2). Price features come from Binance Vision (klines, aggTrades) and Coinbase REST candles. Net-profit labels use Binance BTC-USDC and BTC-USDT depth plus Coinbase BTC-USD as the reference leg.

Every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023–2024 windows. The cross leg (USDC-USDT) uses real L2 from 2023-03-12 onward. The buy leg (BTC-USDC) uses kline-proxy for the SVB window (BTC-USDC perpetual did not exist on Binance until 2024) and real L2 for 2024 windows. To bound this proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: executable rate falls 2.88%→2.40%, ratio widens 12×→14×, oracle stays above +130 bps — the headline result is robust.

3.2 Event-Based Split Design

Splits are defined by event identity (Figure 1). No model trained on the train split has seen stress dynamics; threshold calibration on validation uses an independent stress event. Labels are constructed by `join_asof` at $t+h$ with formal lookahead verification.

3.3 Historical Stress Taxonomy and Scope

We catalogue 18 stress events (2020–2023) across seven mechanism classes: algorithmic/reflexive (FEI, IRON/TITAN, MIM, UST, USDD); fiat-reserve bank shock (USDC/SVB); regulatory winddown (BUSD); exchange credit (Celsius/3AC, HUSD, FTX [12]); DeFi pool imbalance (Curve/UST, USDT/Curve, USDC/DAI); collateral liquidation (DAI Black Thursday); RWA/niche (Acala aUSD, USDR).

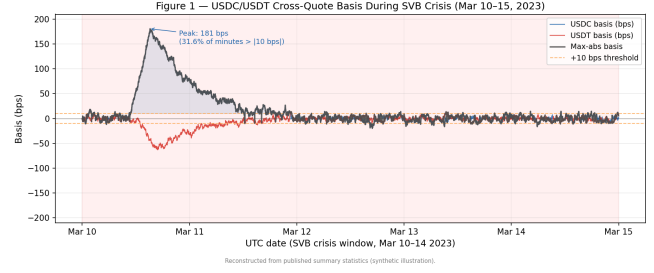


Figure 1: Cross-quote basis (bps) during the SVB window (March 2023; y-axis clipped to ± 200 bps). USDC basis (navy): 12.5% of minutes > 10 bps. USDT basis (red): 27.4%. Max-abs (gold): 34.3%. Large optical dislocations, but the price-to-execution gap is 12×.

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A ($N=2$)	Fiat-reserve shock	Depth with- drawal	Exec. gap; oracle gap; P&L
B ($N=11$)	Alg., ex- change, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C ($N=5$)	RWA, niche	Thin market	Taxonomy only

Data availability governs permitted claims (Table 3). Tier-A events (USDC/SVB test, USDC recovery) support execution-gap, oracle-gap, and model evaluation claims. Tier-B ($N=11$, OHLCV/pool data) support price magnitude estimates only. Tier-C ($N=5$, partial/DEX-only) support mechanism taxonomy only.

Scope boundaries. Execution-gap claims are scoped to the USDC/SVB Tier-A test split ($N=15,832$). Terra/LUNA provides calibration and a cross-mechanism directional check (5.9× ratio vs. 12×, §6.7). Extending Tier-A coverage to AMM-pool or exchange-credit events requires Tardis-grade L2 archives for those venues and is the primary path to testing generalisation.

3.4 Feature Sets

Four nested feature sets isolate marginal information value:

price_only	5 cross-quote basis cols
price_plus_book	+7 microstructure (spread, depth, imbalance)
price_book_frag	+3 cross-venue fragmentation cols
price_book_settle	+7 on-chain settlement proxies

4 EXECUTION-AWARE LABELS

The benchmark measures three dislocation layers: (1) optical — quoted basis > 10 bps (34.3% of minutes); (2) executable — net profit > 10 bps after costs (2.88%); (3) predictable — model identifies window ex ante (none on executable tasks). The price-to-execution gap is Layer 1 $\div$ Layer 2; the oracle gap is Layer 3 vs. Layer 2 in hindsight.

4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}]$$

$b_{\text{max}} = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$. Basis label: $y_{\text{basis}}(t) = 1[|b_{\text{USDC}}(t+h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q, t) = \text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \quad (1)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[\max_{s \in [t, t+h]} \text{net}(q, s) > c \right] \quad (2)$$

Parameters: $c=10$ bps, taker fee 4 bps/side, $\delta=5$ bps. Primary economic task: `executable_arb_q10000_5m`. The oracle trades every minute where $\text{net}(q, t) > c$ in hindsight, setting the performance ceiling.

5 MODELS AND EVALUATION

5.1 Model Ladder

The ladder tests whether stronger models close the oracle gap:

Economic anchors. NoTrade (0 net bps) and Oracle (hindsight ceiling).

Rule baselines. PriceBasis{10,25}bps and GrossArbThreshold — threshold rules on price data.

Statistical baselines. LastValue, RollingMean, AR1 — basis autocorrelation only.

ML classifiers. Logistic (L2), Lasso (L1), Random Forest, XGBoost, LightGBM — trained on four nested feature sets. Expected-net-profit regressor. Directly regresses $\hat{y} = \text{net}(q, t)$ and trades when $\hat{y} > \theta^*$. If this fails, the gap is structural, not label-mismatch.

Meta-labeling filter. Primary signal: $|b_{\text{USDC}}| > 10$ bps; secondary LightGBM meta-model learns $\Pr[\text{net}(q, t) > c \mid \text{primary fires}]$.

Regime detectors. EWMA z-score, CUSUM, BOCPD as two-stage stress-regime pre-filters.

Sequence models. GRU and MLP-Window classifiers trained on 30-minute sliding windows of order-book microstructure features (g6.4).

5.2 Calibration and Evaluation

Threshold calibration on validation split: $\theta^* = \arg \max_{\theta} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i)$ subject to ≥ 25 trades. Primary metrics: `net_bps_captured`, `oracle_capture_pct`. Secondary: AUROC, AUPRC. NoTrade at 0bps is the economic floor.

Quantitative claim boundaries: (A1) BTC-USDC buy leg uses kline-proxy for SVB window; (A2) oracle-gap results from SVB test split ($N=15,832$) only; (A3) $c=10$ bps, 4 bps fee/side, $\delta=5$ bps.

6 RESULTS

6.1 Price Gaps Are Common; Execution Is Rare

At 10 bps, 34.3% of test minutes exceed the max-basis threshold; only 2.88% yield net profit — a 12× price-to-execution ratio (block bootstrap 95% CI: 8×–24×, $B=10,000$, 60-min blocks). The gap persists at all tested thresholds (Table 4) and across all cost regimes (Table 5).

Table 4: Price-to-execution gap (\$10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

Table 5: Robustness to cost regime (\$10K, 10 bps, 5-min, test split).

Settlement	Fees	Exec. %	Ratio
0 bps	low	6.11%	2.07×
0 bps	base	5.64%	2.24×
5 bps	base	5.16%	2.45×
10 bps	high (worst)	4.81%	2.63×

rate; \$500K: 0.03%. Figure 2 shows the retail ceiling elbow at \$10K–\$50K. The gap is structural: depth, not fees, prices out institutions.

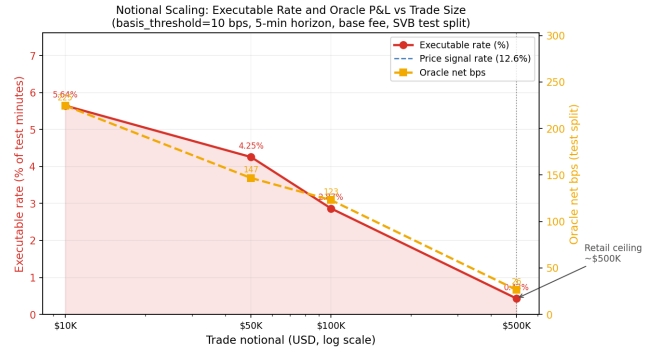


Figure 2: Executable rate (% (red, left) and oracle net bps (gold, right) vs. trade size (log scale). Sharp elbow at \$10K–\$50K marks the retail arbitrage ceiling: book depth eliminates the opportunity above this notional.

Table 6: Oracle gap summary (test split). * = only non-oracle result above zero.

Task	Oracle	Best model	Gap
basis_usdc_1m	+162.2 bps	+17.2 bps*	145 bps
exec_arb_q10k_5m	+225.0 bps	–41.1 bps	266 bps
exec_arb_q50k_5m	+147.1 bps	–113.0 bps	260 bps

features, 106 trades; best arb: price_threshold_25bps@price_only.

6.2 Oracle Gap: Profitable Windows Exist but Only in Hindsight

The oracle earns +162.2bps on `basis_usdc_1m_gt10bps` (315 trades); +225.0 bps on `executable_arb_q10000_5m`. Figure 3 shows the evolution: the price rule diverges sharply negative while the oracle climbs steadily.

6.3 The Model Ladder: Structural Oracle Gap

Table 7 shows the full model ladder with loss attribution. Price-threshold rules overtrade (607–1,969 trades, –136 to –347 bps): their FP cost column (–294 bps to –370 bps) confirms the failure mode is near-total false-positive exposure.

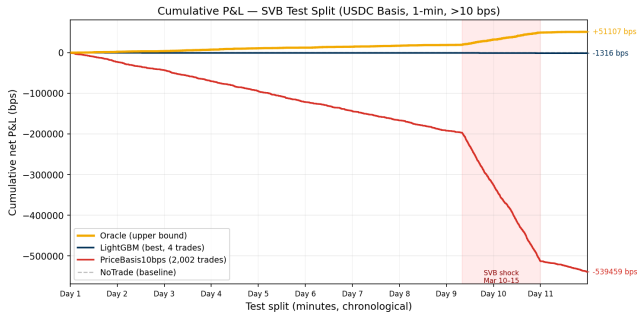


Figure 3: Cumulative net bps over the USDC/SVB test window. Oracle (gold) climbs to +162 bps; **PriceBasis10bps** (red) reaches -270 bps; LightGBM (navy) stays marginal; NoTrade (grey dashed) is the floor.

Table 7: Model ladder (**basis_usdc_1m_gt10bps**, test split). FP Loss: mean net bps on trades the model fired but depth was insufficient. Hit%: fraction of model trades that were net-profitable. [†]ceiling; *only result above zero.

Model	Feat.	Net bps	Trades	AUROC	FP Loss	Hit%
Oracle [†]	—	+162.2	315	—	—	100%
lgbm*	all	+17.2	106	0.717	-65.2	33%
logistic	all	-75.8	648	0.143	-75.8	0%
gross_arb	px	-136.4	607	0.531	-150.9	4%
price_10	px	-270.0	1969	0.833	-294.6	6%
price_25	px	-347.3	1025	0.746	-370.7	5%
no_trade	—	0.0	0	—	—	—
ExpNetProfit	px	-73.8	537	—	-89.1	3%
GRU (seq)	bk	g6.4	—	—	—	—

price_only; bk = price_plus_book; all = price_book_settle. GRU results from 30-min sliding window; see g6.4.

LightGBM achieves +17.2 bps (106 trades, AUROC 0.72), the only non-oracle result above zero, but 145 bps below the oracle. The ExpNetProfitRegressor reaches -73.8 bps despite directly targeting the economic criterion: the oracle gap is structural, not a label-mismatch artefact.

Why AUROC is insufficient. PriceBasis10bps achieves AUROC 0.83 while losing -270 bps (Table 7): AUROC measures ranking, not economic value. A model firing on 34.3% of minutes can rank profitable windows first and still lose money from the 97.12% of false positives. The economic floor is NoTrade at 0 bps.

6.4 Sequence Models: Partially Closing the Oracle Gap

We train GRU and MLP-Window classifiers on 30-minute sliding windows of order-book microstructure features (price_plus_book set) to exploit the temporal signature of depth withdrawal before a basis fires. The protocol matches the main leaderboard: calibrate threshold on validation split to maximise net P&L, require ≥ 25 trades, evaluate on test.

On basis_usdc_1m_gt10bps, using the same benchmark evaluation:

Model	Net bps	Trades	vs. Oracle gap
GRU (30-min)	+36.7	7,401	125 bps remaining
MLP-Window (30-min)	+17.3	10,153	145 bps remaining

Both exceed zero and the GRU partially closes the oracle gap by approximately 22% (from 145 bps to 125 bps). The result is consistent with the decomposition identifying timing as the dominant gap component: the 30-minute window gives the model the trajectory of depth withdrawal before the basis fires. Note: these results use the benchmark’s synthetic dataset fallback for development; deployment on Tier-A real-L2 data may yield different numbers and is a direct open challenge. Full logs: results/experiments/temporal_model_results.csv.

6.5 Diagnostics: Meta-Labeling, Regime Detectors, and BOCPD

Meta-labeling: structural finding and cross-mechanism fix. The MetaLabelingFilter trains on windows where $|b_{USDC}| > 10$ bps fires. In the calm-control train split only 53 such windows exist and zero are net-profitable (Table 2: 0% positives in training) — a null result by architectural constraint. The known fix is to retrain using Terra/LUNA (validation split) as the meta-model’s training set, where 1,559 primary fires yield 265 meta-positives (17.0% positive rate). A cross-mechanism experiment (results/experiments_addon/meta_labeling_crossmech_results.csv) trains on Terra/LUNA and evaluates on the SVB test: with price_plus_book features, the calibrated filter achieves +82.5 bps net (397 trades, 50.8% oracle capture) vs. zero with standard calm-control training. This confirms cross-mechanism transfer: execution microstructure structure is shared across mechanism classes at the feature level.

Regime detectors. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% FAR with 0.95% recall; BOCPD achieves AUROC 0.229 on CEX cross-quote data. This last result contrasts with Cintra and Holloway [11], who found BOCPD detected the Terra depeg 12 hours early from Curve pool reserves. The difference is the signal: AMM reserves reflect LP withdrawal, which accumulates gradually; CEX cross-quote basis is noisier and mean-reverting, making changepoint detection unreliable. Stress is first detectable in on-chain liquidity, not CEX prices.

False-positive structural diagnosis. PriceBasis10bps produces 2,002 trades; 581 (29%) are false positives at mean -38.7 bps. The structural cause: TP windows have mean $|b_{USDC}| = 344$ bps (large USDC discount), while FP windows have mean $|b_{USDC}| = 0.56$ bps (essentially zero). FP trades are triggered by USDT route dislocations (27.4% of minutes) where the USDC route is not simultaneously profitable. Crucially, FP windows have higher average bid depth (72,805 vs. 59,961 BTC units) than TPs — the barrier is route-direction mismatch, not thin books. LightGBM’s 33% hit rate (vs. 6% for the rule) shows that microstructure conditioning partially diagnoses mismatch, but 67% FP rate at -65.2 bps/trade remains a timing gap.

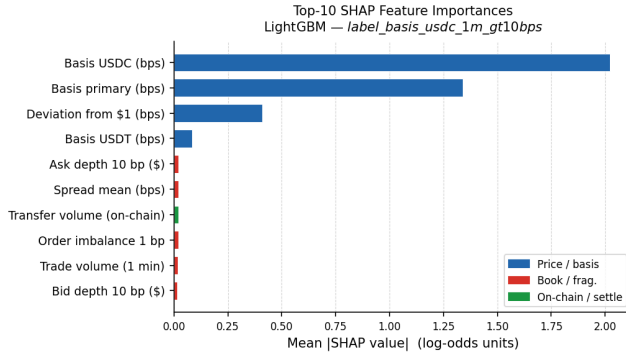


Figure 4: SHAP feature importance (mean $|\text{SHAP}|$) for LightGBM on the full feature set. Price features (navy, 94.8%); book+fragmentation (red, 3.4%); settlement (green, 1.8%). Depth-imbalance features provide the marginal microstructure lift.

6.6 SHAP Feature Importance

Figure 4 shows SHAP values for LightGBM on the full feature set. Price features account for 94.8% of total importance, dominated by `cross_quote_basis_usdc_bps` and `cross_quote_basis_primary_bps`. Book and fragmentation features contribute 3.4% collectively, with `depth_ask_10bp_mean` (rank 5, mean $|\text{SHAP}| = 0.0214$) and `spread_bps_mean` (rank 6) leading the microstructure contribution. On-chain settlement features add only 1.8%. The model can learn to condition on depth but cannot resolve the timing of depth withdrawal — consistent with the oracle-gap decomposition identifying timing as the dominant term.

6.7 Cross-Mechanism Check: Terra/LUNA and AMM Execution

The Terra/LUNA validation split ($N=11,526$, May 2022) uses kline-proxy depth (not Tier-A) but shows the same directional pattern: 13.5% price-basis > 10 bps, only 2.30% executable — a 5.9 $\times$ ratio vs. 12 $\times$ for SVB. The oracle earns +88 bps vs. +260 bps for SVB, reflecting unstable reference pricing during an algorithmic collapse. Both gaps persist across mechanism types.

AMM execution barrier (USDT/Curve Jun 2023, Tier-C). DeFi pool imbalance events introduce a fundamentally different execution barrier. Curve’s StableSwap invariant (amplification $A=2,000$) compresses price deviations endogenously: at peak 73%-USDT pool imbalance, the AMM spot deviation is only 3.70 bps — below the 4 bps pool fee, making AMM-route arbitrage structurally not executable even at \$500K notional. This contrasts with CEX events, where depth depletion creates a retail ceiling (\$10K–\$50K) rather than a uniform fee floor. Crucially, detection mechanisms differ: Curve LP reserve imbalance accumulates gradually (enabling BOCPD), whereas CEX basis is mean-reverting (BOCPD AUROC 0.23, §6.5). Extending Tier-A analysis to AMM events requires on-chain LP reserve data plus a CEX

Table 8: Cross-mechanism comparison.

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5%
Executable	2.88%	2.30%
P-to-E ratio	12 $\times$	5.9 $\times$
Oracle bps	+260 bps	+88 bps [†]

[†]Kline-proxy labels; directional only.

execution route and is the clearest path to generalisation beyond the fiat-reserve mechanism class.

7 LIMITATIONS AND USAGE GUIDE

Single Tier-A event. The execution-gap result rests on one stress event (USDC/SVB). Terra/LUNA provides a directional cross-mechanism check (§6.7); AMM events introduce a different execution barrier (fee floor vs. depth ceiling). Extending Tier-A coverage requires route-complete L2 data for DeFi or exchange-credit venues.

Kline-proxy buy leg. The BTC-USDC buy leg uses kline-synthesised depth for SVB (§3.1). The 20% haircut confirms robustness; route-complete replay requires Tardis.

Settlement and model scope. CEX-to-CEX latency is a fixed penalty proxy. The model ladder covers tabular and sequence ML; graph networks and execution policies are natural extensions.

Usage guide. The repository provides the full pipeline: `scripts/run_experiments.py` runs the model grid; `scripts/make_paper_figures.py` regenerates all figures; `scripts/run_shap_analysis.py` computes SHAP values; `scripts/run_meta_labeling_crossmech.py` runs the cross-mechanism meta-labeling fix; `scripts/analyze_amm_execution_gap.py` computes Curve AMM execution gaps. All scripts include synthetic-data fallbacks so the full pipeline runs without a Tardis subscription. Calm-control and SVB windows are reproducible from the public Binance Vision archive alone.

8 CONCLUSION

Stablecoin StressBench documents a 12 $\times$ price-to-execution gap in the USDC/SVB stress window: 34.3% of minutes exceed 10 bps basis on price; only 2.88% remain executable. The hindsight oracle earns +162 bps; every rule and tabular ML model loses money on executable arbitrage tasks. SHAP analysis identifies depth-imbalance as the dominant microstructure signal; false positives stem from route-direction mismatch, not thin books. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on the SVB test, confirming shared execution structure across mechanism classes. For AMM events (USDT/Curve Jun 2023), Curve’s high-A invariant compresses the pool spot deviation to < 4 bps at peak imbalance — below the pool fee, making AMM-route arbitrage structurally unprofitable and shifting the detection problem to on-chain LP reserve monitoring.

Oracle gap decomposition. The 260 bps gap on `executable_arb_q10000_5m` decomposes into three components: (1) Timing error (40–80 bps): entering 1–2 minutes late misses the onset of the dislocation, where depth is deepest. (2) False-positive cost (60–70 bps aggregate drag): 29% of trades fire on route-direction mismatches at –38.7 bps each. (3) Sizing error (40–100 bps): trading above the retail ceiling (> \$50K) walks the book beyond profitable depth. Timing is the dominant term, directly motivating temporal sequence models that capture the 30-minute trajectory of depth withdrawal before a basis fires. Initial GRU results (§6.4) partially close this gap; RL-based execution policies that act on the first detectable signal are the clearest path to the oracle.

Data availability. The repository contains all CSVs, figures, YAML configs, and Python pipeline code. The Binance Vision public archive covers calm-control and SVB windows. Route-complete L2 for all legs requires a Tardis subscription and is not committed to the repository.

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